

Project Testing Phase Document:

Importing Data Using Transform Maps in ServiceNow

1. Introduction & Testing Objectives

This document outlines the testing strategy, test cases, and validation procedures for the ServiceNow data import and reporting project. The objective is to verify that employee data from external spreadsheets is accurately staged, transformed, and loaded into custom tables while maintaining data integrity, handling duplicates via Coalesce, and correctly rendering reports and dashboards.

Key Testing Goals:

- Verify successful table creation and form layout configurations.
- Validate staging data loading through Import Sets.
- Test Transform Map field mapping and data transformation execution.
- Ensure duplicate prevention and record updates work via Coalesce fields.
- Validate reporting output accuracy and dashboard accessibility.

2. Test Case Matrix & Execution Summary

Test ID	Module / Phase	Test Scenario / Description	Expected Result	Actual Result	Status
TC-01	Custom Table	Create target table Employee Test (u_employee_test) and configure string fields.	Table created successfully with custom fields (Employee ID, Name, Email, Department, Location).	Fields visible on form layout.	Passed

TC-02	Staging Area	Load initial dataset (Sample Spreadsheet.xlsx) via Load Data module.	Import set staging table u_employee_import created with 15 inserts and 0 errors.	State: Complete, Code: Success.	Passed
TC-03	Transformation	Execute Transform Map (Sample Spreadsheet Import) linking source fields to target table.	15 records successfully created in Employee Test table without data truncation.	All 15 rows populated accurately.	Passed
TC-04	Coalesce & Updates	Import modified spreadsheet with new records and updated fields (e.g., email/name modifications).	Existing records updated based on Coalesce key; new unique records inserted; duplicate rows ignored.	Inserts: 2, Updates: 2, Ignored: 0.	Passed
TC-05	Idempotency	Re-upload the exact same spreadsheet to verify duplicate handling.	No inserts or updates performed; all records recognized as duplicates	Inserts: 0, Updates: 0, Ignored: 4.	Passed

			and ignored.		
TC-06	Reporting	Generate Pie Chart report (Employees by Department) and Bar Chart report (Employees by Location).	Reports aggregate data correctly grouped by department and location with correct counts.	Visualizations rendered accurately.	Passed
TC-07	Dashboarding	Create custom dashboard (Employee Analytics Dashboards) and add all three reports as widgets.	Reports successfully added to the dashboard widget grid with correct interactive visibility.	Dashboard displays widgets properly.	Passed

3. Detailed Test Scenarios & Step-by-Step Validation

Test Scenario 1: Initial Data Load & Transform (TC-01 to TC-03)

- **Pre-conditions:** Administrator access to ServiceNow instance, prepared .xlsx file with 15 records.
- **Execution Steps:**
 1. Navigate to **Tables > Create New**, create Employee Test, and configure form layout fields.
 2. Search for **Load Data**, upload Sample Spreadsheet.xlsx, and create staging table u_employee_import.
 3. Create a Transform Map mapping source fields to target table fields via Mapping Assist.
 4. Run the transform process.
- **Validation Criteria:** Verify progress log displays State: Complete and Completion code:

Success. Check the Employee Test table list view to confirm all 15 records are present with correct string values.

Test Scenario 2: Coalesce & Update Validation (TC-04 to TC-05)

- **Pre-conditions:** Initial 15 records loaded; Coalesce set to true on the appropriate identifier field.
- **Execution Steps:**
 1. Modify sample spreadsheet data (change existing employee names/emails, add 2 new rows).
 2. Load the data into the existing staging table (Employee Import).
 3. Run the transform map and inspect **Transform History**.
 4. Re-run the import with the same file without changes.
- **Validation Criteria:** Transform history must show correct metrics (e.g., 2 inserts for new records, 2 updates for modified records). A subsequent re-run with identical data must report 0 inserts, 0 updates, and 4 ignored/skipped rows, proving duplication prevention works correctly.

Test Scenario 3: Report & Dashboard Verification (TC-06 to TC-07)

- **Pre-conditions:** Target table populated with active records; pa_dashboards ACL configured with Admin Override.
- **Execution Steps:**
 1. Create Department Pie Chart and Location Bar Chart reports under **Reports > Create New**.
 2. Create a List report for employee details.
 3. Create the Employee Analytics Dashboard via dashboard configuration.
 4. Share and add all three reports to the dashboard.
- **Validation Criteria:** Charts correctly reflect group counts and percentages. Dashboard loads all widgets without permission errors or rendering delays.

4. Test Sign-Off & Conclusion

- **Testing Status:** Completed Successfully
- **Total Test Cases Executed:** 7
- **Passed:** 7
- **Failed:** 0
- **Defects Logged:** None
- **Conclusion:** The data migration pipeline, transformation rules, Coalesce constraints, and reporting analytics have been thoroughly tested and verified. The solution is validated for production deployment or presentation.